Graded exercises

E1 · Direct application. After determining the set on which it is defined, solve the equation $\ln(2x-1) = \ln(x+3)$.

E2 · Application and justification. Determine the real numbers x for which $\ln x + \ln(x-2)$ is defined, then solve $\ln x + \ln(x-2) = \ln 3$.

E3 · Method to organise. Consider $f(x) = \ln(x^2+1)$ and $g(x) = x\ln x$. State their set of definition and calculate their derivatives.

E4 · Choice of strategy. Determine $\lim\limits_{x\to+\infty} \dfrac{\ln x}{\sqrt{x}}$ and $\lim\limits_{x\to 0^+} x^2 \ln x$. Justify the comparative growth results used.

E5 · Synthesis or proof. By studying a function, prove that $\ln x \leq x-1$ for every real $x > 0$. Do not use the concavity of the logarithm directly.